AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

FIN241

Microfinance

Final Exam VB

Fall 2025-2026

Date: 15 – January – 2026

Number of exam pages: 6 (including this cover page)	Time Allowed: 2 hours

Instructions:

1. Take FIVE (5) minutes to read all instructions carefully before attempting to write.
2. Mobile phones must be in "SWITCHED OFF" mode throughout the examination.
3. This is a CLOSED book examination. NO documents of any kind are allowed.
4. You MUST answer the questions using only the answer booklets provided
5. You MUST return the examination paper together with your answer booklets.
6. Do NOT write your answers in pencil except graphs/sketches
7. Students are NOT allowed to use any type of dictionary or mobile phones during the examinat
8. Students are allowed to use non-programmable calculators only (if needed).
9. The penalty for an ACADEMIC MISCONDUCT in this examination is an F Grade.

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Part A: Solve BOTH Problems (60 Points)

Problem 1: (35 Marks)
You are given the following **Income Statement** and **Balance Sheet** of **Zenith Microfinance**.

SAMPLE INCOME STATEMENT

Account Name	From 1/1/2024 to 12/31/2024	From 1/1/2023 to 12/31/2023
Financial Revenue	4,011,634	2,966,441
Financial Revenue from Loan Portfolio	3,186,634	2,447,556
Interest on Loan Portfolio	2,541,634	1,974,693
Fees and Commissions on Loan Portfolio	645,000	472,863
Financial Revenue from Investments	650,000	449,632
Other Operating Revenue	175,000	69,253
Financial Expense	269,642	209,092
Financial Expense on Funding Liabilities	242,142	195,526
Interest and Fee Expense on Deposits	65,290	55,000
Interest and Fee Expense on Borrowings	176,852	140,526
Other Financial Expense	27,500	13,566
Net Financial Income	3,741,992	2,757,349
Impairment Losses on Loans	114,698	56,956
Provision for Loan Impairment	121,852	80,456
Value of Loans Recovered	(7,154)	(23,500)
Operating Expense	1,976,616	1,455,777
Personnel Expense	1,050,258	1,015,633
Administrative Expense	926,358	440,144
Depreciation and Amortization Expense	125,236	72,555
Other Administrative Expense	275,436	367,589
Net Operating Income	1,650,678	1,244,616
Net Non-Operating Income/(Expense)	678,067	589,621
Non-Operating Revenue	202,369	102,634
Non-Operating Expense	475,698	486,987
Net Income (Before Taxes and Donations)	2,328,745	1,834,237
Taxes	175,236	161,895
Net Income (After Taxes and Before Donations)	2,153,509	1,672,342

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Donations	1,062,563	827,598
Donations for Loan Capital	267,639	364,000
Donations for Operating Expense	1,764,563	1,282,896
Net Income (After Taxes and Donations)	3,216,072	2,499,940

SAMPLE BALANCE SHEET

Account Name	As of 12/31/2024	As of 12/31/2023
ASSETS		
Cash and Due from Banks	675,963	248,698
Trade Investments	2,360,963	5,869,258
Net Loan Portfolio	13,250,558	8,668,369
Gross Loan Portfolio	13,504,693	8,958,369
Impairment Loss Allowance	(254,135)	(290,000)
Interest Receivable on Loan Portfolio	465,000	350,000
Accounts Receivable and Other Assets	481,000	351,896
Other Investments	372,569	372,896
Net Fixed Assets	2,214,824	1,931,856
Fixed Assets	3,065,693	2,439,752
Accumulated Depreciation and Amortization	(850,869)	(507,896)
Total Assets	19,820,877	17,792,973
LIABILITIES		
Demand Deposits	725,896	156,875
Short-term Time Deposits	698,932	525,894
Short-term Borrowings	453,961	100,639
Interest Payable on Funding Liabilities	275,122	177,852
Accounts Payable and Other Short-term Liabilities	575,000	575,000
Long-term Time Deposits	3,505,693	3,505,693
Long-term Borrowings	789,000	689,000
Other Long-term Liabilities	7,023,604	5,730,953
Total Liabilities		
EQUITY		
Paid-In Capital	2,032,317	2,089,000

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Donated Equity	10,195,518	9,367,791
Prior Years	9,228,963	8,562,894
Current Year	966,555	804,897
Retained Earnings	(17,203)	(41,270)
Prior Years	(192,693)	(191,859)
Current Year	135,489	150,639
Reserves	626,642	646,449
Other Equity Accounts		
Adjustments to Equity		
Total Equity	12,797,273	12,062,020

Required: You are asked to analyze the financial strength of Zenith Micro-finance by calculating the following ratios for 2024 and comparing them to the industry averages: **[Note: you should show the formula and every step of your calculation]**
(5 Marks for every ratio)

Ratio	Industry Averages
Portfolio to assets	70.13%
Operating efficiency ratio	51.27%
Portfolio yield	26.37%
Return on assets (ROA)	10.45%
Return on equity (ROE)	15.33%
Debt to equity (leverage)	54.88%
Equity to assets ratio	64.56%

Problem 2:

Four customers of the MFI Amanat have borrowed US$800 each. The loan will be reimbursed in 20 equal weekly installments of $40.

- Customer 1 has regularly reimbursed 10 installments and failed to repay the eleventh.
- Customer 2 reimbursed 5 installments and failed to repay the sixth.
- Customer 3 is paying regularly, with the last payment he made being installment number 15.
- Customer 4 reimbursed 12 installments and failed to repay the thirteenth.

Based on this information, you are asked to calculate the Arrears Rate and the Portfolio at Risk (PAR). Interpret the results. **(25 marks)**

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Part B: Choose 1 out of 2 Questions (40 Points)

Question 3:
Agricultural finance, primarily a subset of rural finance, is dedicated to financing agriculture-related activities such as inputs, production, storage, processing, and marketing of goods? You are required to discuss the challenges of agricultural credit. Support your answer with relevant examples. **(40 marks)**

Question 4:
Explain how FinTech has impacted the cost structure of microfinance institutions (MFIs) in terms of physical network costs and IT infrastructure and storage costs. Discuss how these changes improve efficiency and reduce expenses. **(40 marks)**

***************End of questions***************

Formula Sheet

- APR = (AE + LL + CF + K –Ii) / (1-LL)
- AE = (salaries + rent + depreciation + other administrative expenses) / Average outstanding portfolio
- LL = ((Loan loss reserve year(n-1) + Loan loss provision year(n))- Loan loss reserve year(n) / average outstanding portfolio
- Average equity for cost of funds calculation = Average equity – average net fixed assets
- Forced savings = Average forced savings x rate on saving services
- subsidized loan= Average subsidized loans x rate on commercial loans
- Commercial loan = Average commercial loans x rate on commercial loans
- Equity= Average equity for cost of funds calculation x inflation rate
- CF = (Forced savings + subsidized loan + Commercial loans + Equity) / Average outstanding portfolio. note: the numbers in the numerator are used after applying the market rates
- Ii = Income from other than loan portfolio / Average outstanding loans
- Arrears Rate = Total loan installments past due / Gross portfolio outstanding

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- Portfolio at Risk (PAR) = Total outstanding loan balance on delinquent loans / Gross portfolio outstanding
- Loan Loss rate = total amount written off (in the period) / Average outstanding portfolio (in the period)
- INFLATION ADJUSTMENT = (Average equity – Average net fixed assets) x Inflation rate
- COST OF FUNDS ADJUSTMENT = Average subsidized funds x (commercial interest rate – interest rate paid) OR *(Average subsidized funds x commercial rate) - interest rate paid*
- OSS = operating income / total operating expenses
- FSS = Adjusted operating income / adjusted operating expenses

Portfolio to assets	Gross loan portfolio/assets
Operating efficiency ratio	Operating expense/revenues
Cost per active client	Operating expense/average number of active clients
Borrowers per loan officer	Number of active borrowers/ number of loan officers
Active clients per staff member	Number of active clients/total number of personnel
Portfolio yield	Cash received from interest, fees, and commissions on loan portfolio/ average gross loan portfolio
Return on assets (ROA)	Net income after taxes and before donations/average assets
Return on equity (ROE)	Net income after taxes and before donations/average equity
Nonperforming loans as of 30 days past due (NPL30)	(Nonperforming loans> 30 days + value of renegotiated loans)/gross loan portfolio
Renegotiated loans ratio	Value of renegotiated loans/gross loan portfolio
Write-off ratio	Value of loans written off/average gross loan portfolio
Debt to equity (leverage)	Liabilities/equity
Equity to assets ratio	Total equity/total assets
Liquid ratio (quick ratio or acid-test ratio)	(Cash + trade investments)/ (demand deposits + short-term time deposits +short-term borrowings + interest payable on funding liabilities +accounts payable + other short-term liabilities)
Liquid assets to total assets	Cash/total assets
Loans to deposits ratio	Gross loan portfolio/deposits

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